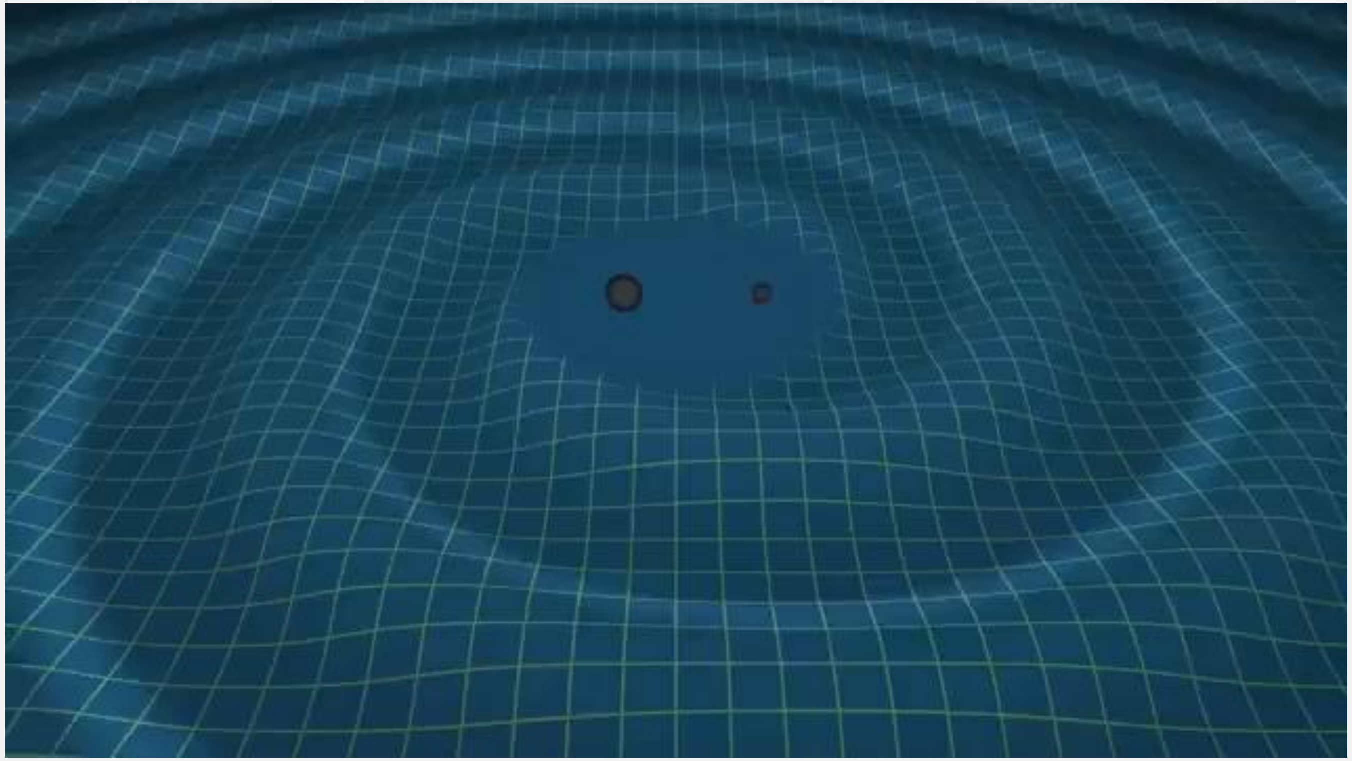
Generate Binary Black Hole Initial Data with Physics-Informed DeepONet

Hilary Wu
Imperial College London
Supervisor: Professor Yang Xiang

Background

- Processing of gravitational wave data
- Traditional numerical relativity (NR) techniques vs. ML techniques
- Focus on non-spinning binary black hole (BBH) systems



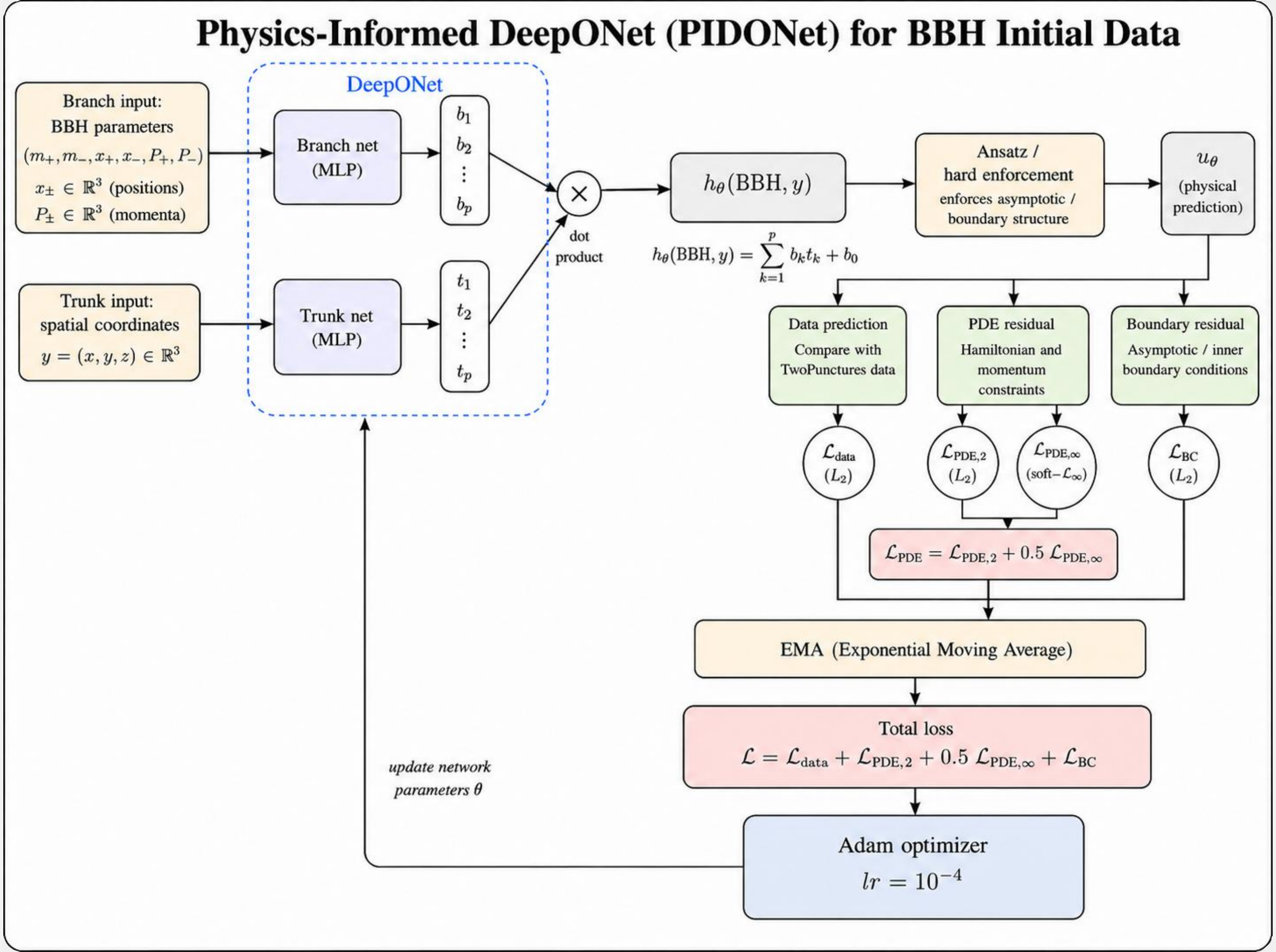
Theory

Einstein's Field Equations:

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}$$

Hamiltonian Constraint Equation:

$$\Delta \mathrm{u} + \frac{1}{8}\left(1 + \sum_{n} \frac{m_n}{2r_n} + \mathrm{u}\right)^{-7} \bar{K}_{ij}\bar{K}^{ij} = 0$$


Physics-Informed DeepONet (PIDONet) for BBH Initial Data
DeepONet
Branch input:
BBH parameters
$(m_+, m_-, x_+, x_-, P_+, P_-)$
$x_\pm \in \mathbb{R}^3$ (positions)
$P_\pm \in \mathbb{R}^3$ (momenta)
Branch net (MLP)
b_1 b_2 $\vdots$ b_p
Trunk input:
spatial coordinates
$y = (x, y, z) \in \mathbb{R}^3$
Trunk net (MLP)
t_1 t_2 $\vdots$ t_p
dot product
$h_\theta(\mathrm{BBH}, y)$
$h_\theta(\mathrm{BBH}, y) = \sum_{k=1}^{p} b_k t_k + b_0$
Ansatz / hard enforcement
enforces asymptotic / boundary structure
u_θ (physical prediction)
Data prediction
Compare with TwoPunctures data
PDE residual
Hamiltonian and momentum constraints
Boundary residual
Asymptotic / inner boundary conditions
$\mathcal{L}_{\mathrm{data}}$ (L_2)
$\mathcal{L}_{\mathrm{PDE},2}$ (L_2)
$\mathcal{L}_{\mathrm{PDE},\infty}$ (soft$-\mathcal{L}_\infty$)
$\mathcal{L}_{\mathrm{BC}}$ (L_2)
$\mathcal{L}_{\mathrm{PDE}} = \mathcal{L}_{\mathrm{PDE},2} + 0.5\,\mathcal{L}_{\mathrm{PDE},\infty}$
EMA (Exponential Moving Average)
Total loss
$\mathcal{L} = \mathcal{L}_{\mathrm{data}} + \mathcal{L}_{\mathrm{PDE},2} + 0.5\,\mathcal{L}_{\mathrm{PDE},\infty} + \mathcal{L}_{\mathrm{BC}}$
Adam optimizer
$lr = 10^{-4}$
update network parameters θ

Setup

Generate BBH initial data for

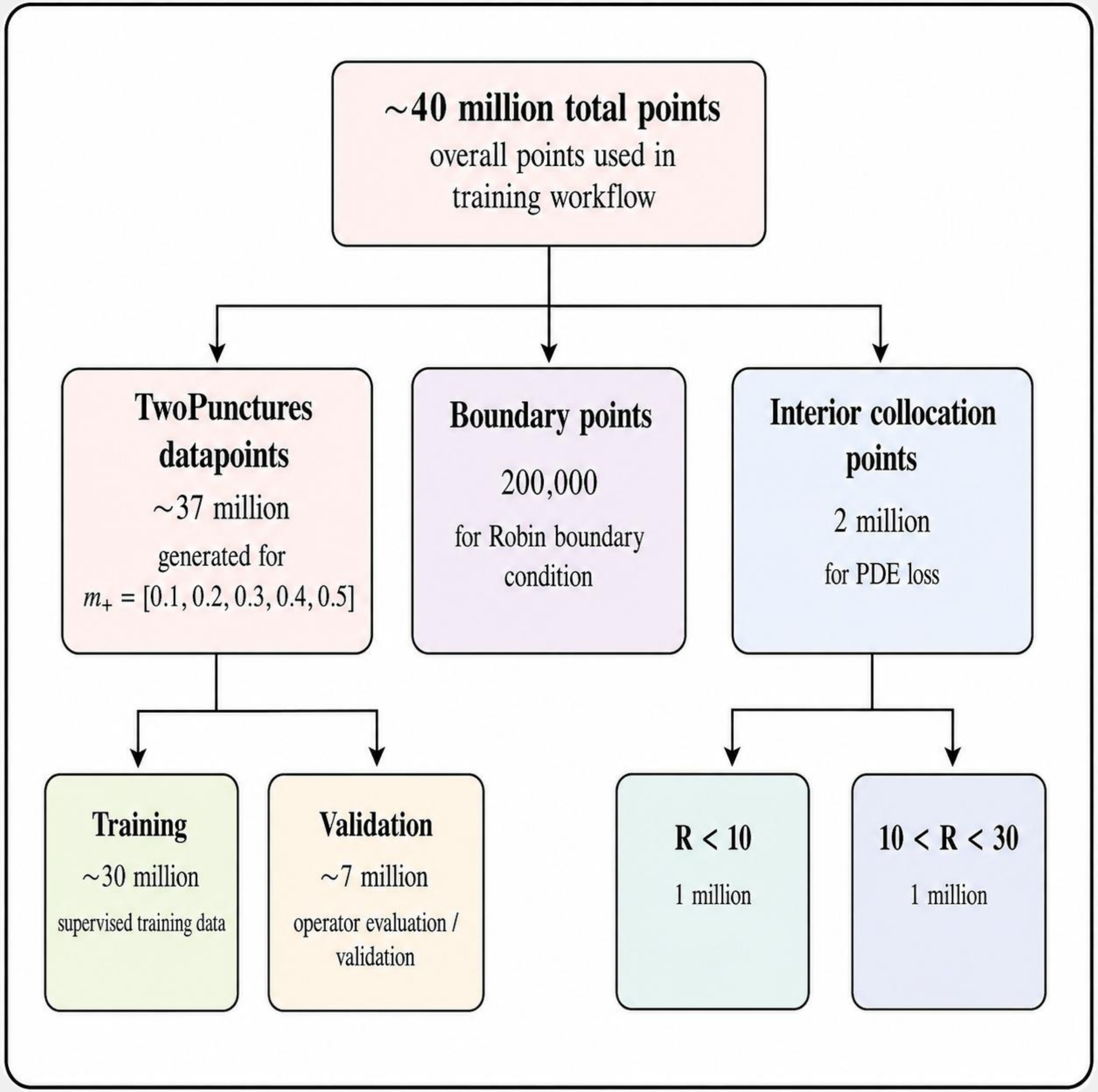
$$m_+ = [0.15, 0.25, 0.35, 0.45]$$

$$m_+ + m_- = 1$$

$$\overrightarrow{X_+} = [3,0,0]$$

$$\overrightarrow{X_-} = [-3, 0, 0]$$

- Spherical domain (R < 30)
- Scale momenta with mass to simulate circular orbit

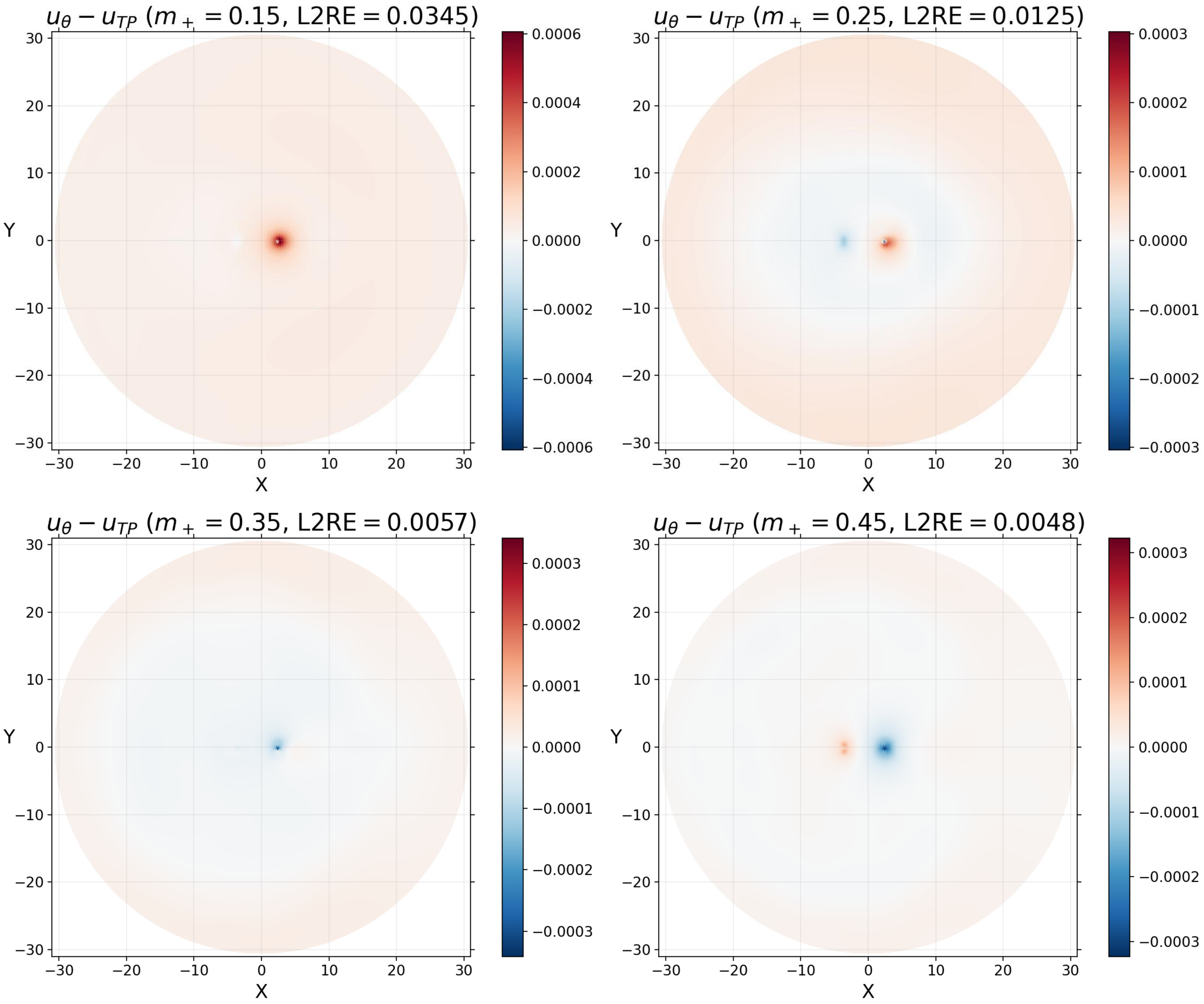


Results

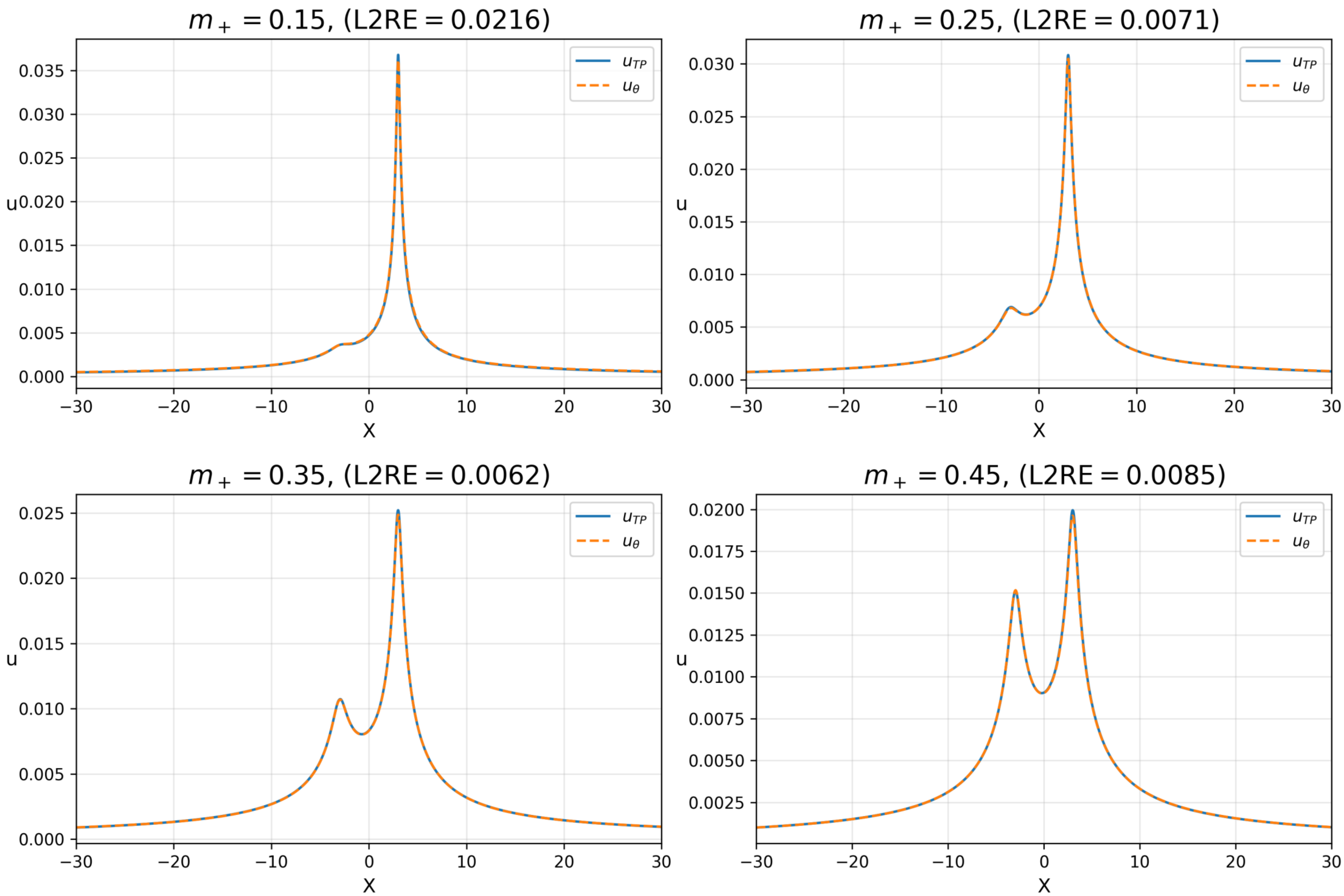
Trained for 2734 epochs (8 hours 20 minutes on A100 GPU)

	L2RE
Validation dataset	0.0202
Test dataset	0.0190
$m_+ = 0.15$	0.0501
$m_+ = 0.25$	0.0204
$m_+ = 0.35$	0.0085
$m_+ = 0.45$	0.0046

$\times$ 2.5 resolution (2D)



$\times$ 25 res (1D)



Conclusion

- Low L2RE on validation (0.0202) and testing (0.0190) datasets
- Lower m_+ values lead to larger deviations from paired u_{TP}
- Generate lower dimensional datasets to up to 25 times initial resolution at 1% average error

Next steps:

- Increase the density of collocation points near individual punctures
- Explore PIDONet's ability to predict u_θ for wider ranges of puncture locations, positions, and momenta
- Extend the branch net to take into account spinning BBH systems

Thank you for listening

Any questions?